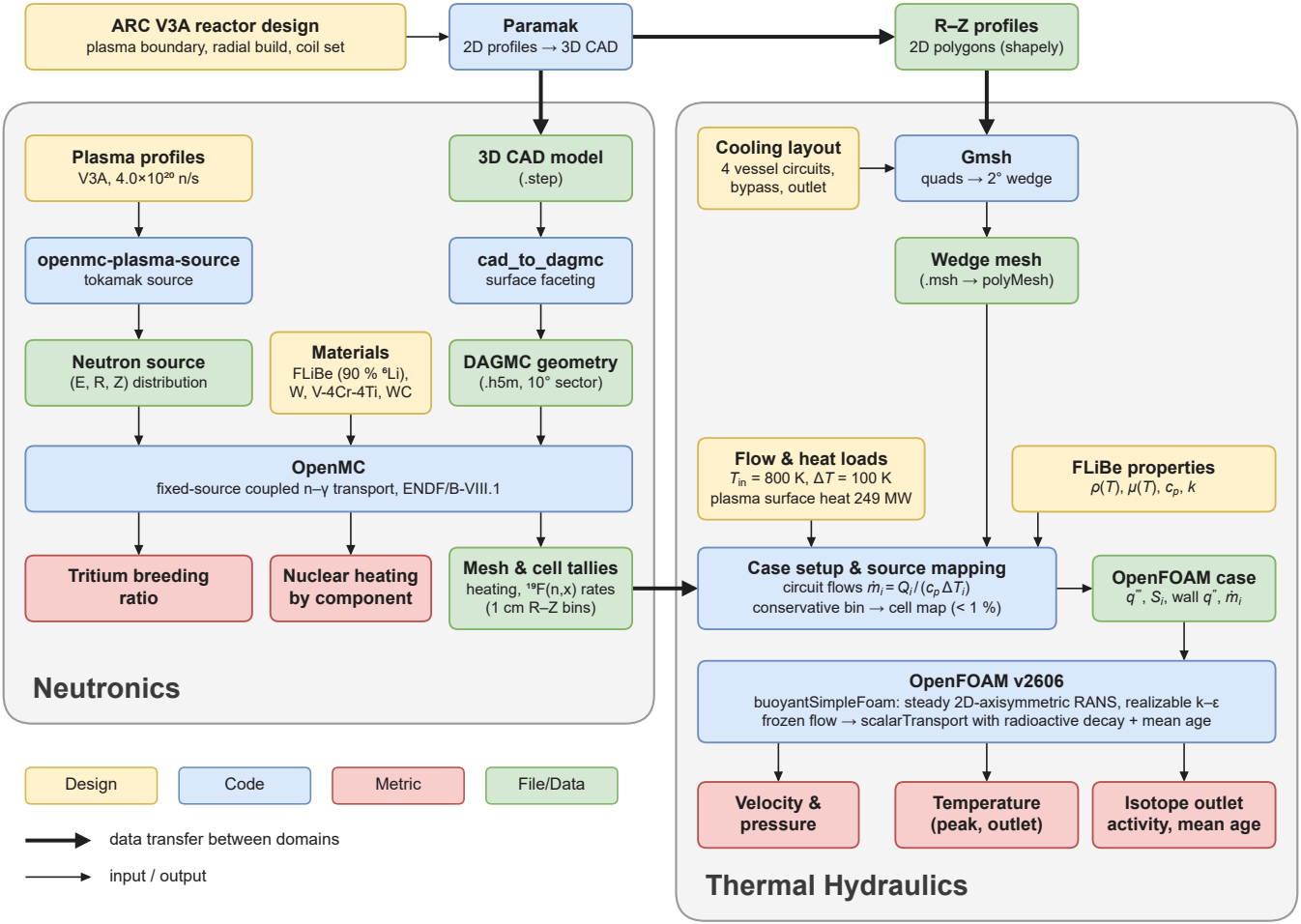

ARC V3A reactor design
plasma boundary, radial build, coil set
Paramak
2D profiles → 3D CAD
R–Z profiles
2D polygons (shapely)
Plasma profiles
V3A, 4.0×10²⁰ n/s
3D CAD model
(.step)
openmc-plasma-source
tokamak source
cad_to_dagmc
surface faceting
Neutron source
(E, R, Z) distribution
Materials
FLiBe (90 % ⁶Li),
W, V-4Cr-4Ti, WC
DAGMC geometry
(.h5m, 10° sector)
OpenMC
fixed-source coupled n–γ transport, ENDF/B-VIII.1
Tritium breeding ratio
Nuclear heating by component
Mesh & cell tallies
heating, ¹⁹F(n,x) rates
(1 cm R–Z bins)
Neutronics
Cooling layout
4 vessel circuits,
bypass, outlet
Gmsh
quads → 2° wedge
Wedge mesh
(.msh → polyMesh)
Flow & heat loads
T_{in} = 800 K, ΔT = 100 K
plasma surface heat 249 MW
FLiBe properties
$\rho(T)$, $\mu(T)$, c_p, k
Case setup & source mapping
circuit flows $\dot{m}_i = Q_i / (c_p \Delta T_i)$
conservative bin → cell map (< 1 %)
OpenFOAM case
q''', S_i, wall q'', $\dot{m}_i$
OpenFOAM v2606
buoyantSimpleFoam: steady 2D-axisymmetric RANS, realizable k–ε
frozen flow → scalarTransport with radioactive decay + mean age
Velocity & pressure
Temperature (peak, outlet)
Isotope outlet activity, mean age
Thermal Hydraulics
Design
Code
Metric
File/Data
data transfer between domains
input / output